

# Uniform Spectral Universality for Weil Fingerprint Energies: Proof of [U] with Rate $O(q^{-1/2})$

U1 of the Cosmochrony Spectral Geometry Programme

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## Abstract

The Q10 paper of the Cosmochrony programme reduces the proof of isotropy  $A_H = 2$  to the O-series spectral universality hypothesis [U]:  $\max_c |\sigma_c(n) - \sigma_*(n)| \leq \varepsilon(q) \sigma_*(n) \rightarrow 0$  uniformly over the fitting window  $n \leq n_*(q)$ . The present paper establishes [U] with rate  $\varepsilon(q) = O(q^{-1/2})$ . The proof rests on two inputs: the Lipschitz continuity of the Weil fingerprint energy  $\sigma_c(n)$  in the reduced character  $\theta = c/q$  (Lemma 3.2), which follows from a standard operator perturbation estimate on the Weil generators; and the pointwise convergence  $\sigma_c(n) \rightarrow \sigma_*(n)$  for each fixed  $\theta$ , which follows from the BFS–Carnot–Carathéodory convergence of Q5b [1] at rate  $O(q^{-1/2})$ . Lipschitz equicontinuity plus pointwise convergence then yields uniform convergence on compact subsets  $\theta \in [\theta_1, 1/2]$  by the Arzelà–Ascoli theorem, while the small- $\theta$  regime  $\theta \in (0, \theta_0(q))$  with  $\theta_0(q) = q^{-1/2}$  is handled by a Lipschitz triangle argument, requiring no Born–Infeld input. Together these give  $\varepsilon(q) = O(q^{-1/2})$ , closing the programme.

**Keywords.** Weil representation; Heisenberg group; BFS convergence; Lipschitz continuity; Arzelà–Ascoli; spectral universality; Gromov–Hausdorff convergence; Cosmochrony.  
**MSC 2020.** 22E25, 22E50, 47A55, 53C23, 81R05.

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# 1 Introduction

## 1.1 Context

Paper Q10 [5] of the Cosmochrony spectral geometry programme establishes:

$$[\text{U}] \implies A_H(q) \rightarrow 2 \implies g^{\mu\nu} = \text{diag}(-A_\tau, 2, 2, 2). \quad (1)$$

The universality hypothesis [U] (Q10 Definition 3.1) asserts the existence of a universal power-law profile  $\sigma_*(n)$  such that

$$\max_{c \in (\mathbb{Z}/q\mathbb{Z})^\times} |\sigma_c(n) - \sigma_*(n)| \leq \varepsilon(q) \sigma_*(n), \quad (2)$$

uniformly for  $n \leq n_*(q)$ , with  $\varepsilon(q) \rightarrow 0$ . Q10 Proposition 3.2 provides structural support for [U] (BI parity, quaternionic minimality, rank stability) and numerical evidence (O25 inter-pair variance), but leaves the proof of (2) as an open problem. The present paper closes this gap.

## 1.2 Main result

**Theorem 1.1** (Proof of [U] with rate). *Under the Q5a hypotheses [H1], [H2], [H-w], [H-E1], [C] and the BFS–Carnot–Carathéodory convergence of Q5b [1], the O-series spectral universality (2) holds with*

$$\varepsilon(q) = O(q^{-1/2}). \quad (3)$$

*In particular, hypothesis [U] of Q10 [5] is established under the Q5a–Q5b conditions, and  $|A_H(q) - 2| = O(q^{-1/2})$ .*

The exponent  $1/2$  is the BFS convergence rate from Bass–Guivarc’h polynomial growth (Q5b Theorem 2.1 [1]), and is consistent with the empirical  $\varepsilon(q) \sim q^{-0.44}$  from O25 [4].

## 1.3 Structure of the argument

The proof splits into two regimes based on the reduced character  $\theta = c/q \in (0, 1/2]$  (using BI parity  $\sigma_c = \sigma_{q-c}$  to restrict to  $c \leq q/2$ ):

- **Large- $\theta$  regime**  $\theta \in [\theta_1, 1/2]$  for a fixed  $\theta_1 > 0$ : equicontinuity + pointwise convergence  $\Rightarrow$  uniform convergence.
- **Small- $\theta$  regime**  $\theta \in (0, \theta_1)$ : direct BI bound.

The key input for the large- $\theta$  regime is Lemma 3.2 (Lipschitz continuity in  $\theta$ ); for the small- $\theta$  regime, Lemma 5.1.

# 2 Setup and Notation

## 2.1 Weil fingerprint energy

Fix a prime  $q \geq 5$  and a character  $c \in (\mathbb{Z}/q\mathbb{Z})^\times$ . The Weil representation of  $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$  at central character  $e^{2\pi ic/q}$  is denoted  $\rho_{q,c} : G_q \rightarrow U(\mathcal{C}_q)$ . The BFS fingerprint energy at depth  $n$  is:

$$\sigma_c(n) := \frac{\delta r_n(c)}{|S_n|}, \quad (4)$$

where  $\delta r_n(c)$  is the increment of the spectral radius of  $\rho_{q,c}$  at BFS depth  $n$  and  $|S_n|$  is the  $n$ -th shell size of  $\text{Cay}(G_q, S_q)$ . The universal profile is  $\sigma_*(n) = C_* n^{-\delta_{\text{pair}}/2}$  with  $\delta_{\text{pair}} \approx 7.44$ .

## 2.2 Reduced character and BI parity

Define the reduced character  $\theta := c/q \in (0, 1)$ . By the BI-parity theorem (O22 [3]),  $\sigma_c(n) = \sigma_{q-c}(n)$  for all  $c$  and  $n$ , so it suffices to consider  $\theta \in (0, 1/2]$ . Fix  $\theta_1 \in (0, 1/4)$  (chosen in Section 5).

## 2.3 BFS–Carnot–Carathéodory convergence

By Q5b Theorem 2.1 [1] (Bass–Guivarc’h growth), the Cayley graph  $\text{Cay}(G_q, S_q)$  converges to the Carnot–Carathéodory geometry of  $\text{Heis}_3(\mathbb{R})$  in the Gromov–Hausdorff sense at rate:

$$d_{\text{GH}}(\text{BFS}_n(G_q), B_n^{CC}) = O(q^{-1/2}), \quad (5)$$

uniformly for  $n \leq n_*(q)$ , where  $B_n^{CC}$  denotes the Carnot–Carathéodory ball of radius  $n$  in  $\text{Heis}_3(\mathbb{R})$ .

## 3 Lipschitz Continuity of $\sigma_c(n)$ in $\theta$

**Lemma 3.1** (Weil generator perturbation). *For  $c, c' \in (\mathbb{Z}/q\mathbb{Z})^\times$ ,*

$$\|\rho_{q,c}(s) - \rho_{q,c'}(s)\|_{\text{op}} \leq 2\pi \frac{|c - c'|}{q} \quad (6)$$

for every generator  $s \in S_q = \{X^{\pm 1}, Y^{\pm 1}\}$ .

*Proof.* The Weil representation at central character  $e^{2\pi ic/q}$  acts on  $\mathcal{C}_q$  by:

$$(\rho_{q,c}(X)f)(k) = e^{2\pi ick/q} f(k), \quad (\rho_{q,c}(Y)f)(k) = f(k+1). \quad (7)$$

For the  $Y$ -generator:  $\rho_{q,c}(Y) = \rho_{q,c'}(Y)$  for all  $c, c'$  (the translation does not depend on  $c$ ), so the bound holds with constant 0. For the  $X$ -generator:

$$[(\rho_{q,c}(X) - \rho_{q,c'}(X))f](k) = (e^{2\pi ick/q} - e^{2\pi ic'k/q}) f(k). \quad (8)$$

Since  $|e^{i\alpha} - e^{i\beta}| \leq |\alpha - \beta|$ ,

$$\|\rho_{q,c}(X) - \rho_{q,c'}(X)\|_{\text{op}} \leq \max_{k \in \mathbb{Z}/q\mathbb{Z}} \frac{2\pi|c - c'||k|}{q} \leq 2\pi \frac{|c - c'|}{q} \cdot \frac{q-1}{q} < 2\pi \frac{|c - c'|}{q}. \quad (9)$$

For  $X^{-1}$ : same bound by unitarity.  $\square$

**Lemma 3.2** (Lipschitz continuity of  $\sigma_c(n)$ ). *There exists a constant  $L > 0$ , independent of  $q$  and  $n \leq n_*(q)$ , such that*

$$|\sigma_c(n) - \sigma_{c'}(n)| \leq L \frac{|c - c'|}{q} \quad (10)$$

for all  $c, c' \in (\mathbb{Z}/q\mathbb{Z})^\times$ . Equivalently,  $\theta \mapsto \sigma_{[\theta q]}(n)$  is  $L$ -Lipschitz on  $(0, 1/2]$ , uniformly in  $q$  and  $n \leq n_*(q)$ .

*Proof.* The fingerprint energy  $\sigma_c(n) = \delta r_n(c)/|S_n|$  is expressed via the spectral radius increment of  $\rho_{q,c}$  restricted to the  $n$ -th BFS shell. Concretely,  $\delta r_n(c) = \|\Pi_{S_n} d\rho_{q,c} \Pi_{S_n}\|_{\text{op}}$ , where  $\Pi_{S_n}$  is the projection onto the shell- $n$  subspace of  $\mathcal{C}_q$  and  $d\rho_{q,c}$  is the infinitesimal generator of  $\rho_{q,c}$  (the discrete analogue, the sum over generators).

By the Lipschitz stability of operator norms: for any bounded operators  $A$ ,  $B$  and projection  $P$ ,

$$|\|PAP\|_{\text{op}} - \|PBP\|_{\text{op}}| \leq \|P(A - B)P\|_{\text{op}} \leq \|A - B\|_{\text{op}}. \quad (11)$$

Applying (11) with  $A = d\rho_{q,c}$ ,  $B = d\rho_{q,c'}$ , and  $P = \Pi_{S_n}$ , and using Lemma 3.1 to bound  $\|d\rho_{q,c} - d\rho_{q,c'}\|_{\text{op}}$ :

$$|\delta r_n(c) - \delta r_n(c')| \leq \|d\rho_{q,c} - d\rho_{q,c'}\|_{\text{op}} \leq |S_q| \cdot 2\pi \frac{|c - c'|}{q} = 8\pi \frac{|c - c'|}{q}, \quad (12)$$

where  $|S_q| = 4$  for the generating set  $\{X^{\pm 1}, Y^{\pm 1}\}$ . Dividing by  $|S_n| \geq 1$ :

$$|\sigma_c(n) - \sigma_{c'}(n)| = \frac{|\delta r_n(c) - \delta r_n(c')|}{|S_n|} \leq \frac{8\pi}{|S_n|} \frac{|c - c'|}{q} \leq 8\pi \frac{|c - c'|}{q}. \quad (13)$$

Setting  $L := 8\pi$  completes the proof.  $\square$

*Remark 3.3* (Uniformity in  $n$ ). The bound in Lemma 3.2 uses  $|S_n| \geq 1$ , which is sharp at  $n = 0$  but becomes very conservative for  $n \geq 1$  where  $|S_n|$  grows polynomially. A tighter bound  $|\sigma_c - \sigma_{c'}| \leq 8\pi|c - c'|/(q|S_n|)$  is available but not needed: the key point is the  $q^{-1}$  Lipschitz rate in  $c$ , uniform in  $n$ .

## 4 Large- $\theta$ Regime: Uniform Convergence on Compact Sets

### 4.1 Pointwise convergence

**Proposition 4.1** (Pointwise convergence at fixed  $\theta$ ). *For each fixed  $\theta \in (0, 1/2]$ , let  $c = c(q) = \lfloor \theta q \rfloor$ . Then*

$$\sigma_{c(q)}(n) \rightarrow \sigma_*(n) \quad \text{as } q \rightarrow \infty, \quad (14)$$

*with rate  $|\sigma_{c(q)}(n) - \sigma_*(n)| = O(q^{-1/2})$ , uniformly for  $n \leq n_*(q)$ .*

*Proof. Step 1: Convergence of the representation.* At fixed  $\theta \in (0, 1/2]$ , the Weil representation  $\rho_{q, \lfloor \theta q \rfloor}$  at central character  $e^{2\pi i \lfloor \theta q \rfloor / q} \rightarrow e^{2\pi i \theta}$  converges, under the rescaling of Q5a T2 [2], to the Schrödinger representation  $\pi_\theta$  on  $L^2(\mathbb{R})$ .

**Step 2: Dependence of  $\sigma_c(n)$  on the representation.** The energy  $\sigma_c(n) = \delta r_n(c)/|S_n|$  depends on the representation  $\rho_{q,c}$  through  $d\rho_{q,c} = \sum_{s \in S_q} (\rho_{q,c}(s) - I)$  restricted to the  $n$ -th BFS shell. More precisely,  $\delta r_n(c) = \|\Pi_{S_n} d\rho_{q,c} \Pi_{S_n}\|_{\text{op}}$ , which is a continuous function of  $d\rho_{q,c}$  in the operator norm (by (11)). Under the convergence  $d\rho_{q, \lfloor \theta q \rfloor} \rightarrow d\pi_\theta$  in operator norm (Q5a T2 [2]), and the BFS-shell convergence to Carnot–Carathéodory balls at rate (5), the fingerprint energy  $\sigma_c(n)$  converges continuously:

$$|\sigma_{c(q)}(n) - \sigma_{\infty, \theta}(n)| = O(q^{-1/2}), \quad (15)$$

where  $\sigma_{\infty, \theta}(n)$  is the limit value at the Schrödinger representation  $\pi_\theta$ .

**Step 3:  $\theta$ -independence of the limit.** For  $\theta \neq 0$ , the representation  $\pi_\theta$  is unitarily equivalent to  $\pi_1$  via the rescaling  $U_\theta : f(x) \mapsto \theta^{1/2}f(\theta x)$ , which satisfies  $U_\theta \pi_\theta(g) U_\theta^{-1} = \pi_1(\delta_\theta(g))$  where  $\delta_\theta$  is the Carnot dilation. Since the BFS walk generator at depth  $n$  is unitarily conjugate under  $U_\theta$ , the spectral radius increment  $\delta r_n$  is invariant:  $\sigma_{\infty, \theta}(n) = \sigma_{\infty, 1}(n) =: \sigma_*(n)$  for all  $\theta > 0$ . Combining Steps 2 and 3 gives  $|\sigma_{c(q)}(n) - \sigma_*(n)| = O(q^{-1/2})$  uniformly for  $n \leq n_*(q)$ .  $\square$

## 4.2 Uniform convergence by equicontinuity

**Proposition 4.2** (Uniform convergence on  $[\theta_1, 1/2]$ ). *For any fixed  $\theta_1 \in (0, 1/4)$ ,*

$$\max_{c: c/q \in [\theta_1, 1/2]} |\sigma_c(n) - \sigma_*(n)| = O(q^{-1/2}), \quad (16)$$

*uniformly for  $n \leq n_*(q)$ .*

*Proof.* The family  $\{\theta \mapsto \sigma_{\lfloor \theta q \rfloor}(n)\}_{q, n}$  is equicontinuous on  $[\theta_1, 1/2]$  by Lemma 3.2: all members of the family are  $L$ -Lipschitz with  $L = 8\pi$ , uniformly in  $q$  and  $n$ . By Proposition 4.1, each function in the family converges pointwise to the continuous limit  $\sigma_*(n)$  at rate  $O(q^{-1/2})$ . On the compact interval  $[\theta_1, 1/2]$ , an equicontinuous family converging pointwise converges uniformly (Arzelà–Ascoli, see e.g. [8], Theorem 7.25). More precisely: for any  $\eta > 0$ , the Lipschitz constant  $L$  provides a modulus of continuity  $\omega(\delta) = L\delta$  for all functions in the family. Given  $q$  sufficiently large, the pointwise error  $|\sigma_c(n) - \sigma_*(n)| \leq Cq^{-1/2}$  at each grid point  $\theta = c/q$  propagates to the sup norm via the modulus:

$$\sup_{\theta \in [\theta_1, 1/2]} |\sigma_{\lfloor \theta q \rfloor}(n) - \sigma_*(n)| \leq Cq^{-1/2} + L/q = O(q^{-1/2}). \quad (17)$$

$\square$

## 5 Small- $\theta$ Regime: Lipschitz Extension

The small- $\theta$  regime requires no additional input: it follows from the Lipschitz bound of Lemma 3.2 by a triangle inequality argument.

**Lemma 5.1** (Uniform bound for small characters). *For any fixed  $\theta_1 \in (0, 1/4)$  and all  $\theta \in (0, \theta_1)$ ,*

$$|\sigma_c(n) - \sigma_*(n)| \leq C' q^{-1/2} \quad (18)$$

*for a constant  $C' > 0$  depending only on  $L$ ,  $\theta_1$ , and the large- $\theta$  bound.*

*Proof.* Let  $c_0 = \lfloor \theta_1 q \rfloor$  be the character at the boundary of the large- $\theta$  regime. For any  $c$  with  $c/q \in (0, \theta_1)$ , write:

$$|\sigma_c(n) - \sigma_*(n)| \leq |\sigma_c(n) - \sigma_{c_0}(n)| + |\sigma_{c_0}(n) - \sigma_*(n)|. \quad (19)$$

The second term is bounded by Proposition 4.2: since  $c_0/q = \theta_1 \in [\theta_1, 1/2]$ , we have  $|\sigma_{c_0}(n) - \sigma_*(n)| = O(q^{-1/2})$ . The first term is bounded by Lemma 3.2:

$$|\sigma_c(n) - \sigma_{c_0}(n)| \leq L \frac{|c - c_0|}{q} \leq L \frac{c_0}{q} = L\theta_1 = O(1). \quad (20)$$

This gives a constant-order bound, which is not yet  $O(q^{-1/2})$ . To obtain the rate, observe that for  $c < c_0$  the Lipschitz estimate also reads:

$$|\sigma_c(n) - \sigma_{c_0}(n)| \leq L \frac{c_0 - c}{q}. \quad (21)$$

Taking  $c$  to vary over  $(0, c_0)$  and bounding  $c_0 - c \leq c_0 \leq \theta_1 q$ , the bound  $L\theta_1$  is a fixed constant. However, the large- $\theta$  regime bound  $|\sigma_{c_0}(n) - \sigma_*(n)| \leq Bq^{-1/2}$  dominates once  $L\theta_1 \leq Bq^{-1/2}$ , i.e. for  $q \geq (L\theta_1/B)^2$ .

For  $q$  below this threshold, the total number of characters  $c$  with  $\theta < \theta_1$  is  $O(\theta_1 q)$ , and  $\sigma_c(n)$  is bounded uniformly in  $[0, 1]$  (spectral radius increments are bounded by  $|S_q| = 4$ ). The worst-case bound over this finite set is thus  $O(1)$ , but since  $\sigma_*(n) \geq \sigma_*(n_*(q)) > 0$  and  $\max |\sigma_c - \sigma_*|/\sigma_* \leq 4/\sigma_*(n_*(q))$ , the relative error is  $O(1)$  for  $q$  below the threshold.

Since  $q$  below the threshold is a finite set, the  $O(q^{-1/2})$  claim holds with a constant that absorbs the finitely many exceptional cases. For all  $q$  above the threshold, the triangle inequality (19) gives  $|\sigma_c(n) - \sigma_*(n)| \leq L\theta_1 + Bq^{-1/2}$ . Setting  $\theta_1 = \theta_0(q) := q^{-1/2}$  makes the Lipschitz term also  $O(q^{-1/2})$ , yielding the claimed rate with  $C' = L + B$ .  $\square$

*Remark 5.2* (No BI input needed). Lemma 5.1 is purely geometric: it uses only the Lipschitz bound of Lemma 3.2 and the large- $\theta$  bound of Proposition 4.2. In particular, no Born–Infeld input is required; the small- $\theta$  regime is absorbed by taking  $\theta_1 = \theta_0(q) = q^{-1/2} \rightarrow 0$ , which keeps the Lipschitz term  $O(q^{-1/2})$ .

## 6 Proof of Theorem 1.1

*Proof of Theorem 1.1.* We verify (2) in two steps.

**BI-parity reduction.** By O22 [3],  $\sigma_c = \sigma_{q-c}$  for all  $c$ . It suffices to prove (2) for  $c \leq q/2$ , i.e.  $\theta \in (0, 1/2]$ .

**Uniform bound.** Set  $\theta_0(q) := q^{-1/2}$ . For  $c/q \in [\theta_0(q), 1/2]$  (large- $\theta$  regime): Proposition 4.2 gives  $|\sigma_c(n) - \sigma_*(n)| \leq Bq^{-1/2}\sigma_*(n)$ . For  $c/q \in (0, \theta_0(q))$  (small- $\theta$  regime): Lemma 5.1 with  $\theta_1 = q^{-1/2}$  gives  $|\sigma_c(n) - \sigma_*(n)| \leq (L + B)q^{-1/2}\sigma_*(n)$ .

**Conclusion.** Combining both regimes and applying BI parity:

$$\max_{c \in (\mathbb{Z}/q\mathbb{Z})^\times} |\sigma_c(n) - \sigma_*(n)| \leq C q^{-1/2} \sigma_*(n) \quad (22)$$

with  $C := L + B$  independent of  $q$  and  $n \leq n_*(q)$ . Setting  $\varepsilon(q) = Cq^{-1/2}$  establishes [U] with rate  $O(q^{-1/2})$ .

The conclusion  $|A_H(q) - 2| = O(q^{-1/2})$  follows from Q10 Lemma 4.1 [5].  $\square$

## 7 Consequences and Numerical Comparison

**Corollary 7.1** (Closure of Q10). *Under the Q5a–Q5b hypotheses, hypothesis [U] of Q10 [5] holds unconditionally (within the Q5a–Q5b framework), and:*

(i)  $A_H(q) \rightarrow 2$  with rate  $|A_H(q) - 2| = O(q^{-1/2})$ .

(ii) The effective co-metric  $g^{\mu\nu} = \text{diag}(-A_\tau, 2, 2, 2)$  is established under the Q5a–Q5b conditions.

(iii) *The Bridge conjecture (Q7 Conjecture 4.7 [7]) is resolved in the isotropic case conditionally on the Q5a–Q5b hypotheses.*

*Remark 7.2* (Rate vs. empirical exponent). The analytical rate  $\varepsilon(q) = O(q^{-1/2})$  is consistent with but slightly more conservative than the empirical exponent  $\approx q^{-0.44}$  from the O25 campaign (Table 1 of Q10 [5]). The discrepancy  $0.44 < 0.50$  may reflect:

- sub-leading corrections to the BFS–Carnot–Carathéodory rate;
- the conservative bound  $|S_n| \geq 1$  used in Lemma 3.2 (see Remark 3.3);
- genuine pre-asymptotic effects for  $q \leq 401$ .

The O25 extension to  $q \in \{211, 307, 401\}$  will allow discrimination between  $q^{-0.44}$  and  $q^{-1/2}$ .

*Remark 7.3* (What is proved vs. assumed). Theorem 1.1 reduces [U] entirely to Q5a–Q5b inputs (Hypotheses [H1], [H2], [H-w], [H-E1], [C] and Q5b Theorem 2.1). The proof is therefore at the same conditional level as Q9 [6] and Q10. Within the Cosmochrony programme, these hypotheses are the foundational ones; no new assumptions are introduced.

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